

3.2 Vertical and Corresponding Angles

Lesson Introduction

 Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.		 Learning Targets	- I can use vertical angles to solve problems. - I can use corresponding angles formed by parallel lines and a transversal to solve problems.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		N/A	
 Essential Questions	- How are corresponding angles formed by parallel lines and a transversal related? - How can you use vertical angles to solve problems?			
 Lesson Narrative	This lesson begins with mapping a line onto another parallel line through a rigid transformation to explore the angle relationship formed by a transversal crossing those lines. Through this rigid transformation, students determine that corresponding angles are congruent, leading to the Corresponding Angles Theorem. The corresponding angle relationships formed from two parallel lines and a transversal are used, along with vertical angles, to determine the measures of other angles formed by two parallel lines, intersected by a transversal.			
 Component Summary	3.2.1 Warm-Up (~5 min.)	Determine a value in a sequence from given shapes (<i>SE p. 133</i>)	3.2.3 Guided Practice (10-15 min.)	Learn to use the Corresponding Angles Theorem to determine unknown values and angles (<i>SE p. 135</i>)
	3.2.2 Exploration (10-15 min.)	Explore corresponding angles formed by parallel lines and a transversal (<i>SE p. 133</i>)	3.2.4 Your Turn! (10 min.)	Use relationships between vertical angles and corresponding angles to solve problems (<i>SE p. 136</i>)
	3.2.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 138</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Transversal - Corresponding Angles - Corresponding Angles Theorem <u>Additional Terms:</u> - Complementary - Supplementary - Congruent		(Optional) Patty Paper	(Optional) Manipulatives or technology to demonstrate angle relationships formed between parallel lines and a transversal.

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may confuse corresponding and vertical angles because they both are congruent to the original angle. - Students may not set the sum of linear pair angles equal to 180°.		- While this lesson does not require the use of patty paper, students may need it to see the translations of angles to show that they are corresponding. - Consider using objects or technology to demonstrate how the corresponding angles are congruent to each other, regardless of the angle by which the transversal crosses the parallel lines. This can be demonstrated using three wooden dowels, pencils, two lines drawn on the board and a yardstick, etc.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Think-Pair-Share Multiple components of this lesson create opportunities for students to discuss their ideas and thinking with their partners. 3.2.2 Exploration and in 3.2.3 Guided Practice are scaffolded so that students are given the opportunity to think about the problems, and then share and discuss them with their partner.		MA.K12.MTR.5.1: Patterns and Structure 3.2.3 Guided Practice question 1b has students using the structure of the problem to connect mathematical concepts. Students will focus on the details of the problem and relate the previously learned concepts of linear pairs and supplementary angles to describe how to determine the measure of an angle from given information. Provide opportunities for students to create a plan to solve the problem.	
 Differentiate Instruction	Consider opportunities in this lesson where students can first identify the angle relationship before finding a missing value. This allows students to structure their thinking when approaching these types of problems, so that with future problems, this type of support is not needed. Create a kinesthetic entry-point in 3.2.2 Exploration by allowing students to use patty paper to explore transformations involving angles formed by parallel lines and a transversal. Allow students to copy one angle and determine which other angles are congruent by moving the copied angle on top of the other angles through a series of rigid motions.			
Lesson Notes:				

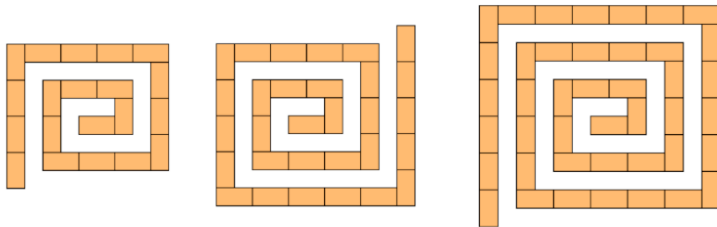
3.2.1 Warm-Up: Shape Sequence

Component Overview: Students are asked to look at a pattern and determine the number of rectangles that will follow in the sequence.



3.2.1 Warm-Up: Shape Sequence

A sequence of three shapes created using rectangles is shown.



1. How many rectangles will have to be added for the next step in the sequence?

14 rectangles

2. How many rectangles will be in Figure 11?

210 rectangles



For students who finish early, consider asking any of the following questions as extensions:

- How many rectangles are in figure 20?
- Can you write an expression that can be used to find the number of rectangles in any figure, n ?

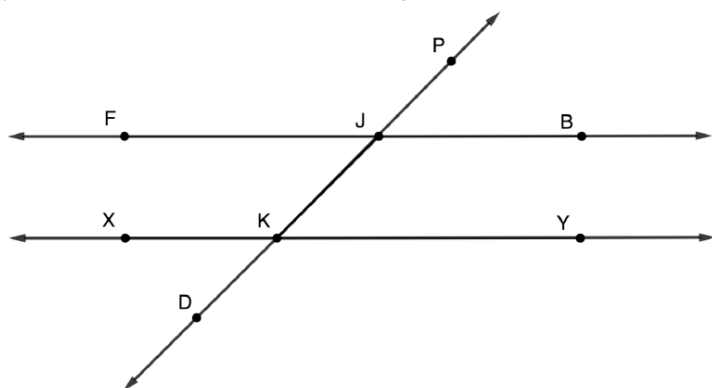
3.2.2 Exploration: Parallel Lines

Component Overview: Students explore relationships when parallel lines are crossed by a transversal, with a focus on corresponding and vertical angles.



3.2.2 Exploration: Parallel Lines

$\overleftrightarrow{FB}$, $\overleftrightarrow{XY}$, and $\overleftrightarrow{DP}$ are shown, where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{FB}$ at point J and where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{XY}$ at point K .



1. Discuss with your partner what rigid transformations can be used to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K .
2. How can you use a translation to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K ?
Translate $\overleftrightarrow{FB}$ down a number of units and to the left a number of units.

In the diagram, $\overleftrightarrow{DP}$ is also known as a transversal, and $\overleftrightarrow{FB}$ and $\overleftrightarrow{XY}$ are parallel lines denoted as $\overleftrightarrow{FB} \parallel \overleftrightarrow{XY}$.



A **transversal** is a line that intersects two or more lines in the same plane at different points.



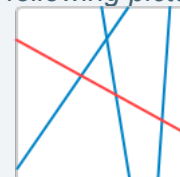
3.2.2 Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group before the 3.2.3 Guided Practice. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.



If students are having difficulties noticing a rigid transformation, providing patty paper can help them better visualize the transformation.



To extend the students' thinking, consider drawing the following picture on the board:



Ask students to discuss with their partner which line(s) would be considered transversal using the definition given.

3.2.2 Exploration: Parallel Lines (continued)

3. Use the translation of $\overrightarrow{FB}$ onto $\overrightarrow{XY}$ so that point J coincides with point K to name the new angles in the chart.

Pre-image	Image
$\angle PJF$	$\angle PKX$
$\angle PJB$	$\angle JKY$
$\angle DJB$	$\angle DKY$
$\angle FJD$	$\angle XKD$

The angles listed in the table above are corresponding angles.



Corresponding angles are angles that are in the same position on two lines in relation to a transversal.

4. Complete the statement.

Given that corresponding angles of parallel lines can be mapped to each other using rigid transformations, it can be concluded that the angles are

- ☐ complementary.
- ☒ **congruent.**
- ☐ supplementary.



Consider providing students with patty paper for this component to add a kinesthetic entry-point.



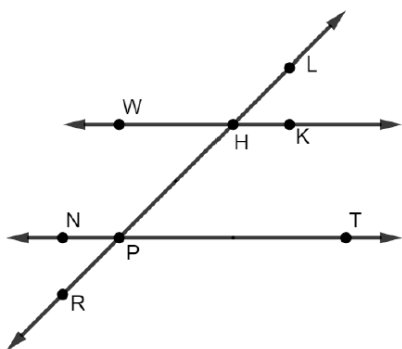
Students may use different points to name the angles in question 3. Have students find a classmate that named the angle differently. Have them discuss why they could both have different answers but still be correct.



Based on the statement from question 4, what can be said about $\angle PJF$ and $\angle JKY$?

3.2.2 Exploration: Parallel Lines (continued)

5. In the figure shown, $\overleftrightarrow{WK} \parallel \overleftrightarrow{NT}$ where $\overleftrightarrow{PH}$ is a transversal.



- a. For each angle, name a corresponding angle and a vertical angle.

	$\angle TPR$	$\angle WHP$	$\angle KHP$	$\angle NPR$
Corresponding	$\angle KHP$	$\angle NPR$	$\angle TPR$	$\angle WHP$
Vertical	$\angle HPN$	$\angle LHK$	$\angle LHW$	$\angle HPT$

- b. Discuss your answers with your partner and make changes, if necessary.



Before students start on question 5a, consider utilizing a Think-Pair-Share asking students to discuss what they know about corresponding and vertical angles. Listen for academic language such as “intersect” and “opposite angles.” Consider having students list similarities and differences between vertical angles and corresponding angles if time allows.

Similarities:

- Formed by intersecting lines
- Congruent angles

Differences:

- Vertical angles are formed by two intersecting lines, and corresponding angles are formed by two parallel lines and an intersecting transversal.

3.2.3 Guided Practice: Angles in Parallel Lines

Component Overview: Students apply the relationships between corresponding angles explored in the previous component to determine the angle measures of angles formed by two parallel lines and a transversal.



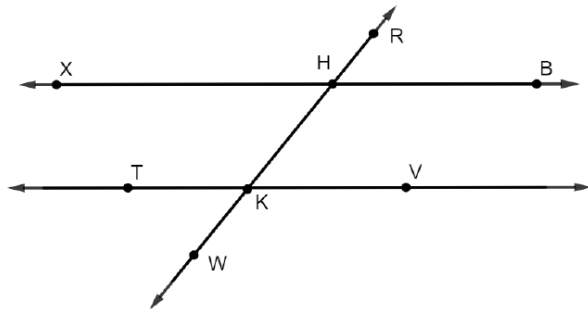
3.2.3 Guided Practice: Angles in Parallel Lines



Corresponding Angles Theorem

If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown such that $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ and $\overleftrightarrow{RW}$ is a transversal.



- a. Given $m\angle RHB = 51^\circ$, determine the measurement of each angle.

$$m\angle HKV = 51^\circ$$

$$m\angle TKW = 51^\circ$$

$$m\angle XHK = 51^\circ$$

- b. Describe how to use $m\angle RHB = 51^\circ$ to find $m\angle XHR$.
Because $\angle XHR$ and $\angle RHB$ are adjacent angles formed by two intersecting lines, then they are a linear pair. The linear pair postulate states that two angles that form a linear pair are supplementary. Supplementary angles sum to 180° by definition so $m\angle XHR + m\angle RHB = 180^\circ$. Therefore $m\angle XHR + 51^\circ = 180^\circ$ and $m\angle XHR = 129^\circ$ by the subtraction property of equality.



For a visual entry-point before students start identifying the angle measure, consider prompting them to highlight all angles congruent to $\angle RHB$. Doing this first will have students focusing on the angle relationships before considering numerical values.



As students share out their answers, listen for and reinforce the use of academic vocabulary already introduced (adjacent angles, intersecting lines, linear pair, supplementary). Once a student shares a correct response, have another student repeat what they said using the correct vocabulary. This move builds toward the justification needed in writing proofs.

3.2.3 Guided Practice: Angles in Parallel Lines (continued)

- c. Explain why knowing $m\angle XHR$ allows all of the angles to be determined.

Once you know $m\angle XHR$, the rest of the angles can be determined using the Vertical Angles Theorem and the Corresponding Angles Theorem.

2. $\overleftrightarrow{MN}$, $\overleftrightarrow{DK}$, and $\overleftrightarrow{PW}$ lie on a plane such that $\overleftrightarrow{MN} \parallel \overleftrightarrow{DK}$ and $\overleftrightarrow{PW}$ is a transversal. $\overleftrightarrow{PW}$ intersects $\overleftrightarrow{MN}$ at point Y and $\overleftrightarrow{DK}$ at point F . $m\angle PYN = (4x + 36)^\circ$, $m\angle NYF = (11y)^\circ$, and $m\angle KFW = 44^\circ$.

- a. Draw a diagram that represents the given information. *Answers will vary.*

Diagram 1

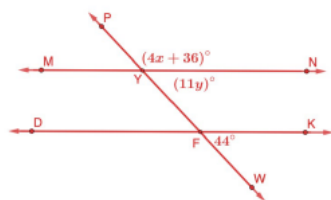
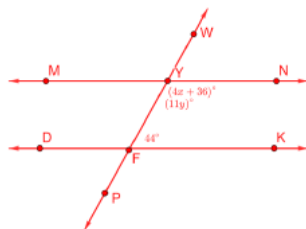


Diagram 2



- b. Determine the value of y .

Answers will vary.

Diagram 1 $y = 4$

Diagram 2 $y = 12\frac{4}{11}$

- c. Determine the value of x .

$x = 25$

- d. What is $m\angle PYM$?

44°



Monitor students as they draw and label the parallel lines and transversal with the appropriate information for question 2a. Consider prompting students to compare their drawings with a partner and resolve any differences prior to proceeding to question 2b - 2d.

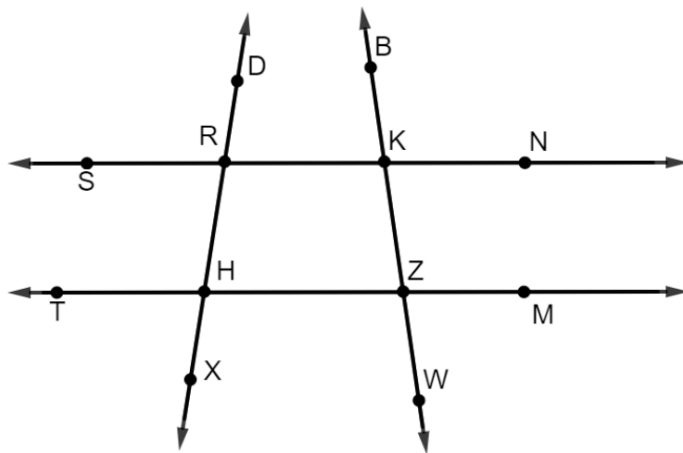
3.2.4 Your Turn!

Component Overview: Students use relationships between vertical angles and corresponding angles to solve questions. Questions include analyzing the reasoning of others.



3.2.4 Your Turn!

1. $\overleftrightarrow{SN}$, $\overleftrightarrow{TM}$, $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ lie on a plane, such that $\overleftrightarrow{SN} \parallel \overleftrightarrow{TM}$ and $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ are transversals. $m\angle BKN = 99^\circ$ and $m\angle KRH = 99^\circ$.



- What is $m\angle KZM$?
 99°
- What is $m\angle ZHX$?
 99°
- What is $m\angle DRS$?
 99°



For a visual entry-point, consider having students first mark all angles on the figure congruent to $\angle BKN$ and then congruent to $\angle KRH$ in a different color, before determining the values of the angle measures.

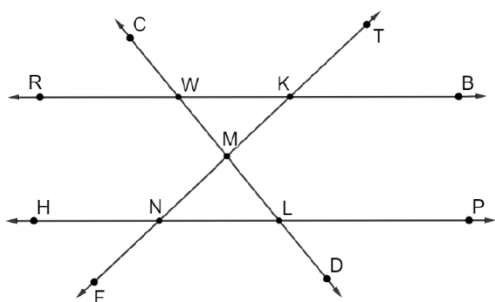
Consider having them share their markings with a partner, explaining how they know the angles are congruent.



For an extension, have students write a justification of their answers. For example, $m\angle KZM$ is 99° because it is a corresponding angle to $\angle BKN$, formed by transversal $\overleftrightarrow{BW}$ intersecting parallel lines $\overleftrightarrow{SN}$ and $\overleftrightarrow{TM}$.

3.2.4 Your Turn! (continued)

2. $\overleftrightarrow{CD}$, $\overleftrightarrow{TF}$, $\overleftrightarrow{RB}$, and $\overleftrightarrow{HP}$ are shown where $\overleftrightarrow{RB} \parallel \overleftrightarrow{HP}$ and $m\angle CWR = 51^\circ$ and $m\angle TKB = 43^\circ$.



Jasmyn's and Drew's justifications of the measure of $\angle CLH$ is shown.

Jasmyn's Answer	Drew's Answer
$m\angle CLH = 51^\circ$ because $\angle CLH$ and $\angle CWR$ are corresponding angles.	$m\angle CLH = 43^\circ$ because $\angle TKB$ and $\angle CLH$ are corresponding angles.

- a. Explain which student you agree with, including your reasoning.

Answers will vary. I agree with Jasmyn because $\angle CLH$ and $\angle CWR$ are corresponding angles. Drew is incorrect because $\angle TKB$ and $\angle CLH$ do not share a transversal, therefore they cannot be corresponding angles.

- b. Find the measure of the following angles.

$$m\angle KNL = \underline{43^\circ}$$

$$m\angle HNF = \underline{43^\circ}$$

$$m\angle MLN = \underline{51^\circ}$$



Consider having students edit Drew's answer to make it correct.

3.2.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



3.2.5 Wrap-Up: Cool Down

1. Draw an example of vertical angles.
Answers will vary.
2. Draw an example of corresponding angles.
Answers will vary.